Practical Assignment Report

Student Activities

Do GeeksforGeeks-Style Software System | Develop likely Linux OS | Generate your own Software development

Module: ITLSE801 - SOFTWARE ENGINEERING

Program: BTech IT

Lecturer: Eng. Ntezirizaza Maurice

Student name: Johnson Mugisha

Registration number: 26RP02147

GitHub: https://github.com/jonishafx

Project link

GitHub profile: https://github.com/jonishafx
This report covers the GeekPath user interface. The interface runs on dummy data in the browser. The Backend folder is not part of this submission.

1. Introduction

This assignment has three practical tasks that build hands-on skills in software development, operating systems, and professional software development. Students work on their own PC, then on a Linux environment, and finally produce a final report. Each task needs evidence such as screenshots, code, recordings if any, and a written report.

This document is the written report for Johnson Mugisha, registration number 26RP02147. The software task is a GeeksforGeeks-style learning interface named GeekPath. It is a frontend-only demo.

2. Reporting requirements for the whole assignment

Each student or group should submit these sections:

- Task 1: software description, screenshots, test results if any, GitHub link.
- Task 2: Linux installation, commands, script, screenshots.
- Task 3: software description, features and innovations.
- Final report including title page, introduction, challenges faced and solutions, review, conclusion, lessons learned, references, and appendices.

3. Task 1 - GeekPath user interface

Software description

GeekPath is a learning-site interface in the style of GeeksforGeeks. It is built with React, TypeScript, and Vite. Pages include home, courses, articles, practice problems, a classroom, a dashboard, certificates, and an admin view.

The Backend folder was removed. The interface does not call a server. Courses, lessons, quizzes, sign-in, progress, and certificates use dummy data saved in the browser.

What changed in the interface

- The header shows the student name, registration number, and a GitHub link.
- The logo and accent color were changed from the original green mark to a GeekPath mark.
- The home page has a sample-data banner.
- The footer shows Johnson Mugisha, 26RP02147, Kigali, and the GitHub link.
- Sign-in, course progress, quizzes, and certificates work from local sample records.

How to run it

From the frontend folder, run npm install and then npm run dev. The site opens at http://localhost:8080/. Demo student login: student@geekpath.dev / password123. Demo admin login: admin@geekpath.dev / admin123. Demo verification code: 123456.

GitHub

Student GitHub profile: https://github.com/jonishafx

4. Challenges and solutions

The first version expected a Node backend and a database. Those calls failed when the server was not running, so course pages and the dashboard could not load. The solution was to drop the backend and keep

every screen on dummy data inside the browser. Sign-in, lesson progress, quizzes, and certificates now update local sample records.

5. Review and conclusion

The submitted software is a user interface only. It demonstrates navigation, course browsing, a classroom, practice problems, and a dashboard without a live server. Johnson Mugisha, registration number 26RP02147, is the author of this report. The GitHub profile for this work is https://github.com/jonishafx.

6. Lessons learned

- A user interface can be demonstrated with sample data before a server exists.
- Browser storage is enough for progress in a classroom demo.
- A report should name the student, the registration number, and the GitHub link on the title page.

7. References

- React documentation: https://react.dev
- Vite documentation: https://vite.dev
- GitHub profile: https://github.com/jonishafx